

DAY 170/365 · PART 1 OF 3

Decision Trees

The flowchart that learns from data.

Part 1: what it is and how it predicts

Part 2: how it picks questions

Part 3: a full worked example

You already use decision trees

“Should I carry an umbrella?”

Is it cloudy? → No → leave it.

Cloudy? → Yes → Is rain forecast? → Yes → take it.

A decision tree is that same chain of yes/no questions, but the model learns the questions from data instead of a person writing them.

Classification

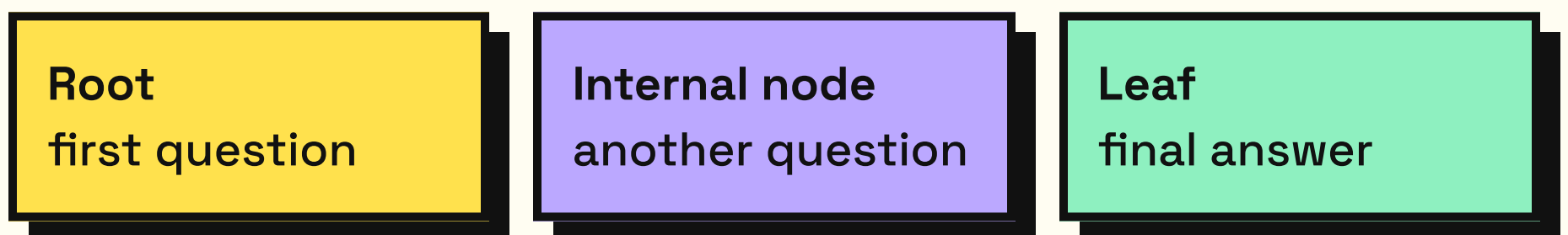
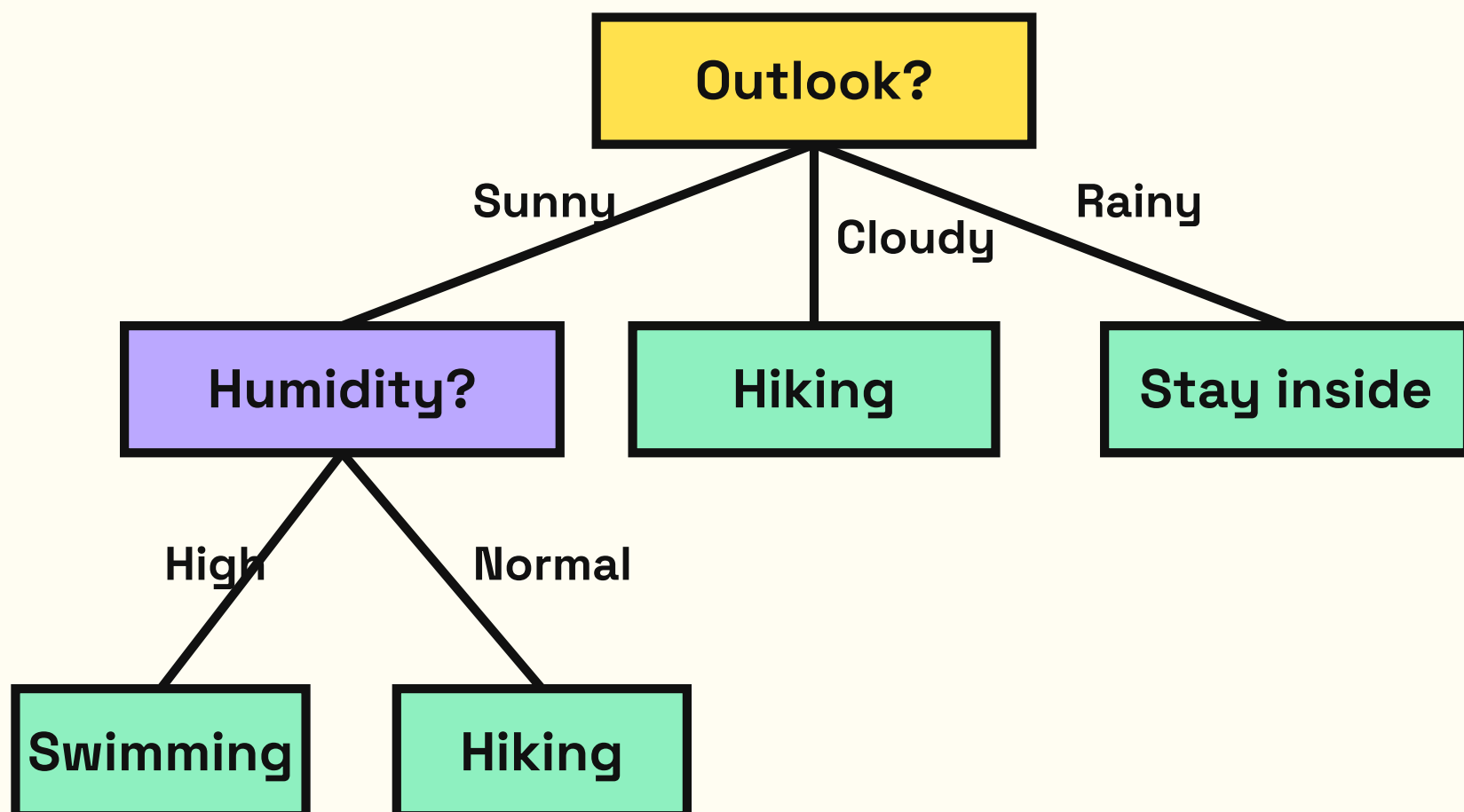
Predicts a category, e.g. buy or not buy.

Regression

Predicts a number, e.g. a car price.

It's supervised learning, so it learns from labelled examples, just like my car price models.

Anatomy of a tree



The lines are branches, one for each possible answer.

Making a prediction = walking the tree

Today is sunny and the humidity is high. What should we do?

1 Start at the root: Outlook?

2 Take the Sunny branch. It leads to Humidity?

3 Take the High branch.

4 You reach a leaf: Swimming 

Try it: Cloudy goes straight to a leaf, Hiking. Rainy goes straight to Stay inside.

Every path is a readable rule

Say a shop wants to guess whether a customer will buy:

IF income > 50,000 **AND** age > 30 **AND**
bought before
THEN predict Purchase

IF income $\leq$ 50,000
THEN predict No Purchase

No hidden maths. You can read the tree out loud and explain it to a manager.

This is why trees are loved for interpretability. In my car price series, I had to dig through coefficients to explain a model. A tree explains itself.

How is a tree built?

1 Put all the data at the root.

2 Pick the best question to split it.

3 Divide the data into groups by the answer.

4 Repeat on each group, if it's still mixed.

5 Stop when a group is pure and label it.

Pure means everyone in the group has the same answer, like a box with only red balls. A mixed box needs another question.

Why use them, and the catch

Good

- Easy to read
- Works for categories and numbers
- Needs little data prep

Catch

- Can grow too deep
- A deep tree memorizes the training data

That catch is the same overfitting story from my Day 168 post. A tree that keeps asking questions until every training row is perfect is like Student B, who memorizes the exam paper.

We'll fix that in Part 3.

TAKEAWAYS

A decision tree is a flowchart that learns its own questions.

Root, branches and leaves. Prediction means walking from the top to a leaf.

Build it by splitting again and again until groups are pure.

Part 2 tomorrow: how does it choose the best question? Entropy, information gain and Gini.